LINEAR REGRESSION

In [2]:
```
import pandas as pd
df=pd.read_csv(r"C:\Users\Admin\Desktop\student_info.csv")
df
```

Out[2]:

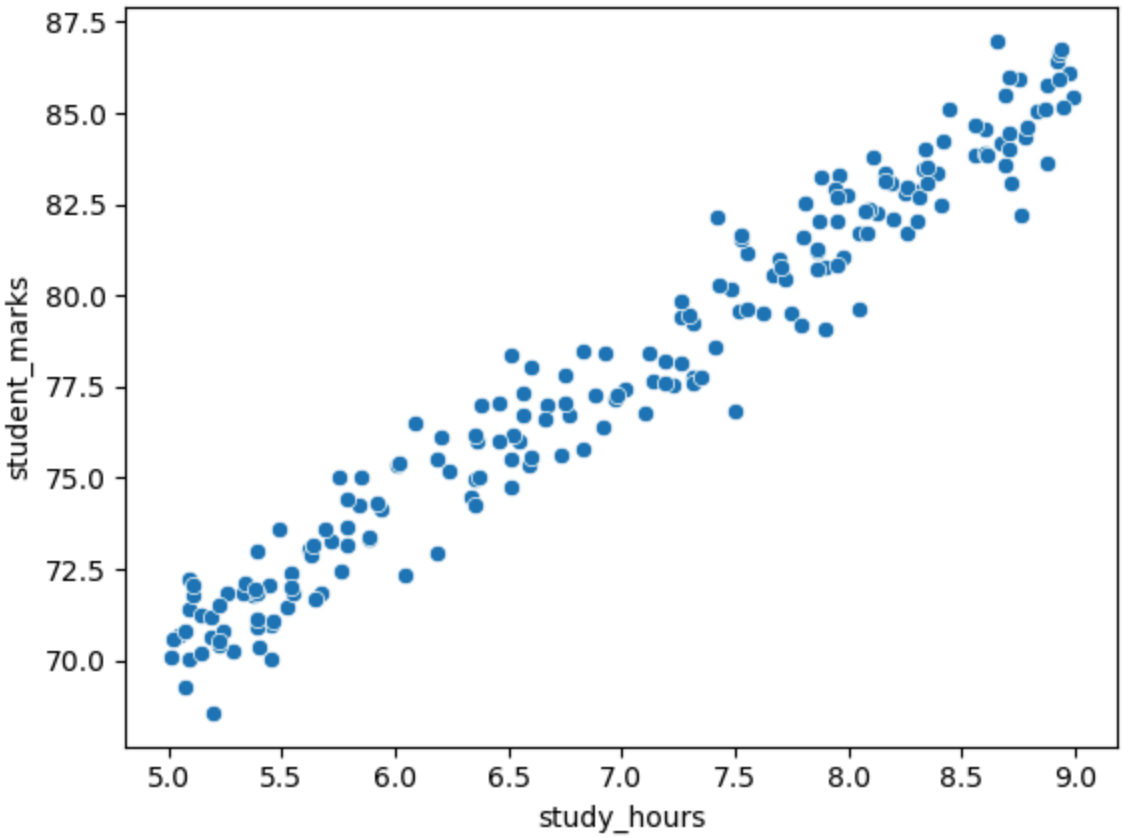
	study_hours	student_marks
0	6.83	78.50
1	6.56	76.74
2	5.67	71.82
3	8.67	84.19
4	7.55	81.18
...	...	...
194	7.53	81.67
195	8.56	84.68
196	8.94	86.75
197	6.60	78.05
198	8.35	83.50

199 rows × 2 columns

In [9]:
```
# DATA VISUALIZATION
import seaborn as sn
```

In [11]:
```
sn.scatterplot(x="study_hours",y="student_marks",data=df)
```

Out[11]:
```
<Axes: xlabel='study_hours', ylabel='student_marks'>
```



In [12]:
```
# LINEAR REGRESSION
# STEP 1: ASSIGNING VARIABLES(INDEPENDENT/DEPENDENT)

# INDEPENDENT VARIABLE
x=df[["study_hours"]]

# DEPENDENT VARIABLE
y=df[["student_marks"]]
```

In [13]:
```
# STEP 2:IMPORT TRAIN TEST SPLIT
from sklearn.model_selection import train_test_split
x_train,x_test,y_train,y_test=train_test_split(x,y,test_size=0.3)
```

In [15]:
```
print("total data: ",df.shape)
print("x_train data: ",x_train.shape)
print("x_test data: ",x_test.shape)
print("y_train data: ",y_train.shape)
print("y_test data: ",y_test.shape)
```

```
total data:  (199, 2)
x_train data:  (139, 1)
x_test data:  (60, 1)
y_train data:  (139, 1)
y_test data:  (60, 1)
```

In [16]:
```
# STEP 3: APPLY LINEAR REGRESSION
from sklearn.linear_model import LinearRegression
model=LinearRegression()
model
```

Out[16]:
```
LinearRegression
▶ Parameters
```

In [17]:
```
# STEP 4 :FIT TO DATA
# Use only train(x_train,y_train)
model.fit(x_train,y_train)
```

Out[17]:
```
LinearRegression
▶ Parameters
```

In [18]:
```
# CHECK ACCURACY
model.score(x_test,y_test)
```

Out[18]:
```
0.9524771238185038
```

In [21]:
```
# PREDICTION OF NEW DATA
# IMPORT NUMPY FIRST(DATA MUST IN ARRAY FORM)
import numpy as np
data=np.array([[5]])
data
```

Out[21]:
```
array([[5]])
```

In [22]:
```
prediction=model.predict(data)
prediction
```

```
C:\Users\Admin\anaconda3\Lib\site-packages\sklearn\utils\validation.py:2749: UserWarning: X does not have valid feature names, but LinearRegression was fitted with feature names
  warnings.warn(
```

Out[22]:
```
array([[70.1475188]])
```

In [ ]:
```

```